

The Word-Complexity Ladder: A Friendly Guide

Companion notes to “A Machine-Verified Rung Zero and a Mahler-Value Reduction of the Automati

June 12, 2026

1 The diary of an orbit

Play the Collatz game from any number: halve if even, triple-and-add-one-then-halve if odd. Alongside the numbers themselves, keep a *diary*: write 1 each time the number was odd, 0 each time it was even. The diary of 3 reads 1100010101... — after five entries the orbit reaches the $1 \rightarrow 2 \rightarrow 1$ loop, and the diary settles into the repeating pattern 1010... forever. Diaries of settled orbits always end in a repeating pattern.

Two facts from the earlier papers in this series set the stage. First, **the diary determines the number**: no two starting numbers ever write the same infinite diary. (Each new entry reveals one more binary digit of where you started — after T entries you know the start modulo 2^T .) Second, **a number that escapes to infinity must keep its diary at least 63.09% ones forever** — the critical density $\log 2 / \log 3$ that runs through this whole subject. Below that share of odd steps, the halvings win and the orbit falls.

So here is a strategy for attacking the divergence half of the conjecture, one class of diaries at a time:

The ladder. Sort all possible diaries by how complicated they are: perfectly repetitive ones at the bottom; then diaries a simple machine could write; then more inventive ones; with truly patternless diaries at the top. Prove, rung by rung, that *no actual number writes a diary that simple*. Every rung is a finite, honest theorem. The conjecture’s divergence half is the (unreachable-for-now) top.

2 Rung zero: repeating diaries belong to settled orbits

The floor of the ladder is now a machine-checked theorem: *a diary that eventually repeats belongs to a bounded orbit*. Why: if the diary from day s onward equals the diary from day $s + P$ onward, then the two orbit values on those days have *identical infinite diaries* — and since diaries determine numbers, the values are equal. The orbit literally loops, so it can never have been heading to infinity. Every step of this argument is formalized in Lean (no axioms, no gaps), including the bookkeeping identity that powers everything else below.

*Written in collaboration with Claude (Anthropic). These notes explain the ideas; the paper and the Lean files contain the proofs. The Collatz conjecture remains open.

An escaping number, then, must improvise forever — its diary can never settle into a repeat. The next rung up asks: could the improvisation at least be *mechanical*?

3 Rung one: diaries written by a copying machine

Here is the simplest kind of aperiodic diary. Take the two rules

$$1 \rightarrow 110, \quad 0 \rightarrow 011,$$

start with 1, and repeatedly rewrite every symbol by its rule:

$$1 \rightarrow 110 \rightarrow 110110011 \rightarrow \dots$$

Each pass triples the length and the text stabilizes into one infinite word $w = 110110011110110011011110110\dots$ — never repeating (the paper gives a four-line proof), yet generated by a two-line program. Words like this are called *automatic*. Note its density of ones: exactly $2/3 = 66.7\%$, comfortably above the critical 63.09%. As far as the density theory is concerned, this diary is *allowed* to belong to an escaping number. Does it?

By fact one, at most one number could own this diary. Which number? The bookkeeping identity of the earlier papers answers exactly: the candidate owner is the value of the series

$$x = -10 + \frac{5}{9} F\left(\frac{8}{9}\right), \quad \text{where } F(z) = w_0 + w_1 z + w_2 z^2 + w_3 z^3 + \dots$$

— the “diary polynomial” of infinite degree, evaluated at the innocent-looking fraction $8/9$. (Why 8 and 9? Three game steps multiply the number by roughly $\frac{3 \cdot 3}{2 \cdot 2 \cdot 2}$ when two of them are odd — the 8 counts the halvings in a block of three steps, the 9 counts the triplings. The fraction $8/9$ is the substitution rule, seen by arithmetic.)

One catch, and it is the heart of the matter: the series must be summed *2-adically* — in the number system where a number is small when it is divisible by a high power of 2 (the natural habitat of Collatz diaries, which are read off by repeated halving). The same series also sums happily as ordinary real numbers — to $-6.5701\dots$, visibly not a positive integer — but that tells us nothing: the real sum and the 2-adic sum of the same series are genuinely different numbers. It is the 2-adic value that would be the escaping integer.

The whole rung in one sentence. The infinite word w copies itself under the rule — so its diary polynomial satisfies the exact self-similarity equation

$$F(z) = \frac{z + z^2}{1 - z^3} + (1 - z^2) F(z^3),$$

and the only number that could own the diary is $-10 + \frac{5}{9} F(8/9)$ summed 2-adically. So: *if $F(8/9)$, summed 2-adically, is not a fraction, then no whole number writes this diary* — one rung of the ladder hangs on the irrationality of one exotic number.

4 A question with a famous shape

“Self-similar function, evaluated at a nice point: can the value be rational?” is a century-old genre of question. Kurt Mahler invented a method for it in 1929, and the modern

theory (Nishioka’s book is the standard reference; Adamczewski and Faverjon’s recent work in its sharpest form) handles exactly equations like the one above. Indeed, for our F the classical theory already delivers a verdict at the *real* number side: $F(8/9)$ summed as real numbers is provably transcendental — not just irrational, but as far from a fraction as a number can be. The hypotheses are checked in the paper: the word never repeats, so F is a genuinely transcendental *function* (a lovely 1906 theorem of Fatou does this step), and the self-similarity equation is of precisely Mahler’s kind.

What rung one needs is the *same* conclusion for the *same* function at the *same* point — but summed 2-adically. The ingredients of Mahler’s proof are not specifically about real numbers, and analysts have transposed his method to other number systems before; but a published theorem in exactly the needed form has not yet been located. That is the paper’s punchline: an honest, well-posed question handed to the right specialists.

5 Why the lazy route provably fails

Couldn’t one just say: the series converges *fast* 2-adically, and numbers that admit too-fast rational approximation are irrational (Liouville’s classic trick)? The paper computes exactly how fast: cutting the series after N terms approximates the value with quality exponent

$$\frac{3 \log 2}{2 \log 3} = 0.9464 \dots$$

— and Liouville-style arguments need quality strictly above 1. Worse (or better): for *any* candidate diary in the allowed band, the quality works out to (critical density)/(diary’s density), which is below 1 precisely because escaping diaries must beat the critical density. The same constant that creates the divergence problem seals off the cheap proof. Crossing from 0.946 to beyond 1 is exactly what Mahler’s auxiliary-polynomial machinery was built to do — speed alone won’t, structure must.

6 The fairness check

A recurring discipline in this series: any method claiming progress on $3x + 1$ must *not* accidentally “work” for the $5x + 1$ game, where orbits really do seem to escape (start at 7 and watch). The ladder passes. For $5x + 1$ the same construction evaluates the same diary polynomial at $8/25$ instead of $8/9$, and the Mahler argument would prove: no whole number writes *this exact mechanical diary* under $5x + 1$ either. That is consistent with real escape — the actual $5x + 1$ escapees write messy, patternless diaries (about 49% ones), nothing like a copying machine’s output. The method excludes specific orderly diaries; it cannot fake a proof that everything settles. That is what soundness looks like here.

7 What this does and does not achieve

Honestly stated: even a full success at rung one would exclude only the most orderly aperiodic diaries — a countable, measure-zero family — and the conjecture would remain open above it. The earlier no-go theorem in this series proves the top of the ladder cannot be reached by

any argument with bounded memory; the ladder is a way of making clean partial progress, not a secret staircase to the summit.

What the paper does deliver: the floor of the ladder machine-verified; the second rung converted — by exact, mostly machine-checked arithmetic — into a single, classical-shaped question about one 2-adic number; a proof that the question genuinely cannot be settled by elementary approximation, with the obstruction quantified ($0.946 < 1$) and traced back to the problem’s own critical constant; and a fairness certificate that the method cannot overclaim. Each piece is checkable: the formal parts by running Lean, the numerical parts by a one-minute script.

8 Summary in four sentences

Every Collatz orbit writes a binary diary of its parities, the diary determines the orbit, and escaping orbits must keep their diaries above 63.09% ones forever — so excluding diaries class by class, ordered by complexity, is a ladder of honest theorems toward the divergence half of the conjecture. Rung zero is machine-verified: repeating diaries belong to settled orbits only. Rung one — diaries written by the simplest copying machines — reduces exactly to whether one self-similar function’s value at $8/9$, summed 2-adically, is irrational: the real-number version is already transcendental by Mahler’s classical method, the cheap approximation route provably falls short by the critical-density margin, and the 2-adic case is a well-posed open question for transcendence theory. The conjecture remains open — but one of its rungs now has the shape of a century-old, well-understood kind of problem.